

REVIEW ARTICLE

Considerations on activity assay discrepancies in factor VIII and factor IX products

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Abstract

New modified coagulation factor VIII (FVIII) and factor IX (FIX) products have been designed to improve the treatment of individuals with hemophilia A and B by increasing the interval between dosing. Although these FVIII and FIX molecules have been structurally modified to improve the circulation time, the changes have also influenced their behavior in functional assays in comparison with traditional plasma-derived or recombinant coagulation factors. The assignment of potencies for these products can be problematic because discordance in factor activity values between the commonly used one-stage clotting and chromogenic substrate assays is often observed. Discrepancies in potency assay values also exist when different assay kits and reagents are used in the same assay type. Ideally, all FVIII and FIX products should be calibrated against the World Health Organization (WHO) International Standards (IS) because the assignment of potencies in international units (IU) helps maintain treatment tradition and meaningful references for manufacturers, patients, and clinicians. The discrepant measurements, attributed to the modified structural and functional properties of these products, are manifested in their lack of commutability with the WHO IS for FVIII or FIX. Herein, we discuss the considerations upon which an assay is chosen for potency assignment and postadministration monitoring of a new factor product, which include the validity of the assay calibrated with the IS, the meaning of the potency values in IU, standards of care for patients, clinical relevance between the assigned potency value and recovery value from clinical laboratories, and patient safety.

KEYWORDS

factor IX, factor VIII, hemophilia A, hemophilia B, hemostasis

1 | INTRODUCTION

Coagulation factor VIII (FVIII) and factor IX (FIX) play pivotal roles in hemostasis, and their congenital deficiency leads to the X-linked bleeding disorders hemophilia A and B, respectively. These

conditions are managed by prophylactic and on-demand regimens using plasma-derived or recombinant FVIII and FIX products.¹ Both regimens are based on the observation that the severity of bleeding and associated arthropathy are directly related to the factor levels of the individual.² The goal of prophylaxis is to maintain trough factor activity levels at least 1% to 5% of normal, which has been shown to dramatically reduce the frequency of bleeding episodes and damage

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to the joints.³⁻⁶ On-demand treatment targets much higher peak factor levels, as appropriate for the severity of the respective bleed (e.g., 20%–40% for minor bleeds and 50%–100% for major bleeding episodes).¹

Correct dosing with these products is critical to achieve factor levels that can effectively maintain hemostasis and, simultaneously, prevent rare but potentially devastating thrombosis when excessive product is administered. Appropriate dosing is assured by the accurate assignment of factor potency of the product, relevant assessment of factor recovery after product administration, and meaningful correlation of the pharmacokinetics (PK) of the product determined in clinical trials with the treatment of individuals using the product postlicensure.

2 | ROLE OF THE INTERNATIONAL STANDARD

To support the licensure and use of coagulation factor products and diagnosis of hemophilia, all factor activity assays are calibrated against the World Health Organization (WHO) International Standards (IS). Two types of WHO IS have been established for FVIII and FIX activity: IS of the factor in plasma, and the other in factor concentrate.⁷ The plasma IS is made from a pool of normal plasma, and the concentrate IS is made from purified FVIII or FIX. The factor activities in these IS are labeled in international units (IU). One IU is defined as the amount of factor activity found in 1 ml of pooled normal plasma (i.e., 1 IU/ml is the normal level of the coagulation factor activity in plasma). Because the potency assignment of factor concentrates using plasma IS as a reference revealed poor assay robustness, IS for factor concentrates were developed to improve the accuracy and reliability of potency assignments for purified factor products.^{8,9} The WHO IS for FVIII and FIX concentrates were initially calibrated against their respective WHO plasma IS. Subsequent WHO IS were calibrated against their predecessor IS (e.g., the 5th WHO IS for FVIII plasma was calibrated against the 4th IS for FVIII plasma¹⁰; and the 6th WHO IS for FVIII concentrate was calibrated against the 5th WHO IS for FVIII concentrate).⁹ Thus, the plasma IS provides a universal standard for the diagnosis of abnormalities in coagulation factor activity and monitoring of treatment with factor concentrates, whereas the concentrate IS aids in the potency assignment of factor products and the calculation of a safe and effective dose.¹¹ The plasma and concentrate IS are important tools to facilitate the interpretation and comparison of factor activity measurements while reducing intra- and inter-assay variabilities.

To compare the factor activity levels generated from different clinical laboratories, the local factor activity assay should be traceable to the plasma IS.¹²⁻¹⁴ This can be achieved by using a reference material derived from a normal plasma pool calibrated against the respective plasma IS. For historical reasons, clinical laboratories usually report FVIII and FIX results as a percentage of the normal level, with the understanding that 100% of this “normal” level corresponds to exactly 1 IU of factor activity in 1 mL of plasma.

Likewise, the potencies of FVIII and FIX concentrates are labelled in IU referenced to their respective concentrate IS, and thus linked to the plasma IU. However, universal labelling of products in IU does not imply that the factor concentrate products will demonstrate similar pharmacokinetics or efficacy in the clinic.¹⁵ Indeed, traceability of product unitage to a single plasma IS should not be construed to mean biological or biochemical equivalency between the factor protein in normal plasma and the protein in the recombinant or plasma-derived factor concentrates.¹¹ By design, the manufacturing processes for plasma-derived and recombinant proteins select for a more homogeneous distribution of protein isoforms (e.g., by removing protein variants with different charges, sizes, or binding properties than the desired protein product). At the same time, certain manufacturing steps may inadvertently modify the protein (e.g., viral inactivation treatment). As a result, recombinant coagulation factor concentrates are usually less heterogeneous than the proteins in plasma, but with different posttranslational modifications (e.g., glycosylation patterns).

The decision on whether the IS for FVIII or FIX concentrates was made from a plasma-derived or recombinant material was determined by the results of the collaborative studies, and consultation with and recommendation from the WHO Expert Committee on Biological Standardization. All previous factor concentrate IS were derived from plasma, except for the 6th WHO IS for FVIII concentrate, which was recombinant and used between 1998 and 2003. This practice may have reflected the prevalence of the use of plasma-derived factor concentrates at the time of the IS development. With the introduction of multiple new recombinant FVIII and FIX concentrates and the concomitant phasing out of plasma-derived products from the market, the use of recombinant candidate materials in future collaborative studies for IS would seem reasonable.

On the one hand, unmodified recombinant materials may be more suited for the calibration of assays for unmodified recombinant products, as evidenced by the improved interlaboratory agreement when three recombinant FIX products were calibrated against the recombinant candidate versus the plasma-derived candidate.¹⁶ On the other hand, a recombinant IS may result in further divergence of the WHO concentrate IS from the WHO plasma IS, and the original definition of the international unit as the amount of factor activity found in one ml of pooled normal plasma. Ultimately, the decision on the source for future concentrate IS will be made based on the results of the collaborative studies, and consultations with and recommendation from the Expert Committee on Biological Standardization.

3 | COAGULATION FACTOR ACTIVITY ASSAYS

Both the one-stage clotting (OC) and chromogenic substrate (CS) assays are routinely used to measure FVIII and FIX activity by clinical laboratories.^{13,16} Presently, the OC assay, available since 1953, monitors the intrinsic (contact) coagulation pathway by measuring the time required for clot formation using the activated partial thromboplastin time (APTT) test.¹⁷ The OC assay is the most commonly

used factor activity assay in the world. In this assay, factor-deficient plasma is first mixed with test plasma and two APTT reagents, a contact pathway activator and procoagulant phospholipids. Coagulation is then initiated by adding calcium to the mixture, and the clotting time is recorded and compared with a standard for the test factor.¹⁸ The OC assay is subject to variability in the quality of the factor-deficient plasma and clotting process. OC assay biases arise from the differences between the multiple brands of APTT reagents, which contain contact activators and procoagulant lipids of different origins and quality. Importantly, these APTT reagents are widely used (and often optimized) for other diagnostic purposes¹⁹ (e.g., for the detection of lupus anticoagulant and overdosing with unfractionated heparins rather than for the robust measurement of FVIII and FIX activity). Therefore, some widely used APTT reagents may show suboptimal sensitivity to low levels of FVIII and FIX or unreliable responses to certain types of recombinant FVIII and FIX products.

The CS assay, developed in the 1980s, uses purified components of the coagulation pathway and removes the need for the plasma matrix and clot formation.^{13,20,21} The CS assay measures the activity of FVIII or FIX via the production of FXa and is performed in two stages. In the FVIII activity assay, the test plasma is mixed with FIXa, phospholipids, and, in some kit versions, purified thrombin, to allow for FVIII activation and formation of the FIXa:FVIIIa intrinsic tenase complex. Similarly, in the FIX activity assay, test plasma is mixed with FXIa, to allow for FIX activation, and FVIII with phospholipids to form the intrinsic tenase complex. Second, in both FVIII and FIX assays, the added FX is then activated to FXa by the intrinsic tenase complex, and the FXa level is measured with a synthetic chromogenic substrate for FXa. When the CS assay is performed properly, the FXa level is proportional to the activity of the test factor. Variabilities between the CS assay kits can come from differences in reagents (e.g., FXIa, FIX, and FX can be sourced either from human or animal plasma) and procedure (e.g., time of activation phase).

4 | ASSAY DISCREPANCIES FOR CONCENTRATES OF FVIII AND FIX

Despite the use of a concentrate IS as a common reference standard, differences in potencies have been reported for multiple factor products between the CS and OC assays, and between OC reagents or CS assay kits.^{11,22-26} These discrepancies can result in disagreements between the potency value assigned by the manufacturer of the product and the activity determined in patient blood or a reconstituted vial in a clinical laboratory. Importantly, test variability or incorrect assay implementation alone cannot explain these discrepancies because similar discrepancies are reported by unrelated laboratories. Thus, consistent biases in reporting the potencies of certain factor products by clinical laboratories and product manufacturers may bring into question the applicability of a universal IU in practice.

Historically, reports of discrepant CS and OC assay results have been known since purified FVIII concentrates became available,²⁷⁻³⁰ although assay issues may have resulted in few dosing or efficacy

concerns. One notable incident was reported for a B-domain-deleted (BDD) recombinant FVIII product, which rendered about a 50% higher potency value by a CS assay than the OC assay.³¹⁻³⁴ The clinical trials that provided the data to support this product's safety and efficacy were performed with subjects being dosed and monitored based on CS-derived FVIII values. The product was licensed using the CS assay for potency assignment.³⁵ Shortly after licensure, cases of lack of effect of the product, as evidenced by unexpected breakthrough bleeds during prophylaxis, were reported.³⁶ A subsequent meta-analysis suggested that this BDD-FVIII product was less effective for prophylaxis as compared to full-length FVIII (FL-FVIII).³⁷ Initially, the lack of effect was attributed to the differences in PK between the FL-FVIII and BDD-FVIII. Subsequent side-by-side comparative studies³⁸ suggested that this was not the case, with the caveat that bioequivalence established from PK data cannot necessarily be extrapolated to suggest comparable clinical safety or efficacy between the BDD-FVIII and other products.¹⁵ Alternatively, it can be argued that even a very small shortening of the product half-life can account for reduced efficacy and a 30% increase in the usage of BDD-FVIII product compared with FL-FVIII products.³⁹

Interestingly, the manufacturer of the affected BDD-FVIII product has also developed a clinical laboratory reference standard, but its labeled activity was later shown to be incorrectly determined. A multilaboratory collaborative study⁴⁰ confirmed that the manufacturer had overestimated the potency of the material, and suggested that there were issues with the manufacturer's CS method. To correct the potency labeling issues, the amount of FVIII protein per 1 IU in all product vials has had to be increased by 20% since 2003.^{40,41} The amount of protein was increased again by 12% in 2006 to align potencies with the new WHO IS for FVIII concentrate.⁴² In 2008, after the product had undergone significant manufacturing changes, the potency was re-assigned to an OC-based unitage in the United States and Canada, whereas the European version of the same product continued to be tested using a CS assay. As a result of these regional differences in assay requirements for FVIII products, 1 IU of this BDD-FVIII product marketed in North America is equivalent to 1.38 IU of the same product distributed in Europe.^{39,43}

In recent years, several modified recombinant FVIII and FIX products (see Tables 1-3) have been developed, expanding the therapy options for people with hemophilia. For many of these new products, discrepancies in potency assignments are observed between the OC and CS assay systems despite the use of the same IS with each assay. As expected, the magnitude of assay discrepancies appears to depend on the types of OC and CS reagents used by clinical laboratories.⁴⁴⁻⁵⁰ It remains a challenge for all stakeholders to reconcile the assay results and resolve this situation.¹¹

5 | STRATEGIES FOR CHOOSING A POTENCY ASSAY

Currently, no one single assay for FVIII and FIX products can be considered as a reference assay by all clinical laboratories.¹¹

TABLE 1 Marketed FVIII products, factor modifications and factor activity assay information (information from the DailyMed, a publicly available database which contains labeling submitted to the US FDA by companies⁶²)

INN	Product	Company	Source/Modification	Potency Assay	Clinical Laboratory Assay Recommendations
Octocog alfa	Advate	Shire	Recombinant	OC	OC
Rurioctocog alfa pegol	Adynovate	Shire	Recombinant/PEGylated	OC	OC
Lonoctocog alfa	Afstyla	CSL Behring	Recombinant/B-domain deleted/Single Chain	CS	OC or CS. Multiply OC value by 2
n/a ^a	Alphanate	Grifols	Plasma derived	OC	No recommendation
Efmoroctocog alfa	Eloctate	Bioverativ	Recombinant/B-domain deleted/Fc-fusion	CS	OC or CS
Turoctocog alfa pegol	Esperoct	Novo Nordisk	Recombinant/B-domain deleted/glycoPEGylated	CS	CS or "appropriate" OC
Damoctocog alfa pegol	Jivi	Bayer	Recombinant/B-domain deleted/PEGylated	CS	CS or "selected" OC or central laboratory
n/a ^a	Koate DVI	Kedrion	Plasma derived	OC	No recommendation
Octocog alfa	Kogenate FS	Bayer	Recombinant	OC	OC
Octocog alfa	Kovaltry	Bayer	Recombinant	CS	OC or CS
n/a ^a	Hemofil [®] M	Shire	Plasma derived	OC	No recommendation
n/a ^a	Humate-P	CSL Behring	Plasma derived	OC	No recommendation
Turoctocog alfa	NovoEight	Novo Nordisk	Recombinant/B-domain deleted	OC	OC or CS
Simoctocog alfa	Nuwiq	Octapharma	Recombinant/B-domain deleted		No recommendation
Susoctocog alfa	Obizur ^b	Shire	Recombinant/B-domain deleted/porcine sequence	OC	OC
Octocog alfa	Recombinate	Shire	Recombinant	OC	OC
n/a ^a	Wilate	Octapharma	Plasma derived	OC	No recommendation
Moroctocog alfa	Xyntha/ Refacto	Pfizer	Recombinant/B-domain deleted	OC/CS ^c	OC

Note: For product dosing recommendations, please refer to the package inserts.

CS, chromogenic substrate; US FDA, United States Food and Drug Administration; INN, International Nonproprietary Name; n/a, not available; OC, one-stage clotting.

^a INN is not available for a plasma-derived product.

^b Indicated only for adults with acquired hemophilia A.

^c CS assay calibrated with an OC assay-calibrated reference standard.

Furthermore, this problem is exacerbated by different regional practices (e.g., European regulations have specific requirements on the type of assays used to assign potencies for certain factor products, whereas the United States and Canada do not). All currently licensed FVIII and FIX products are assigned potency values by either the CS or OC assay even when there is disagreement between the two, and the reasoning for choosing either CS- or OC-based potency labeling is not always articulated. In this report, we discuss the rationale and strategies for selecting between the CS and OC assays for factor potency assignment, and issues related to assay harmonization, clinical relevance, and patient safety.

5.1 | Assay validity

Hubbard et al⁵¹ delineated consensus regulatory science recommendations on potency labeling of FVIII and FIX concentrates for manufacturing and marketing purposes, as well as guidance to

physicians on the dose to be used for treatment that would correlate with recovery data measured in clinical laboratories. The paper also provided a decision tree for the potency assignment of new factor products.⁵¹ Briefly, new FVIII and FIX products need to be assessed for the analytical validity of the potency assignment (e.g., dilutional parallelism) against the current WHO IS using both the CS and OC assays. If the product shows analytical validity in both assays, the calculated potencies are compared. If the values are different, further agreement between the manufacturer and regulatory agency is needed on how to report the assay-specific potency in IU. If the product displays a lack of parallelism to the WHO IS in both assays, the results can be reported in "product-specific units" or in mass units (e.g., milligrams). One such example is a FVIIIa-mimetic bispecific antibody product, for which deviations from parallelism with the WHO IS for FVIII concentrate can be demonstrated in various FVIII activity assays.⁵² However, the potency assignment is not performed because this antibody product is dosed by mass. Another example of assay-specific issues is

TABLE 2 Marketed FIX products, factor modifications and factor activity assay information (information from the DailyMed, a publicly available database which contains labeling submitted to the US FDA by companies⁶²)

INN	Product	Company	Source/Modification	Potency Assay	Clinical Laboratory Assay Recommendations
n/a ^a	Alphanine SD	Grifols	Plasma-derived	OC	No recommendation
Eftrenonacog alfa	Alprolix	Bioerativ	Recombinant Fc-fusion	OC	OC, not kaolin-based
Nonacog alfa	BeneFix	Pfizer	Recombinant	OC	OC
Albutrepenonacog alfa	Idelvion	CSL Behring	Recombinant albumin-fusion	OC	OC but over- and underestimation reported
Trenonacog alfa	Ixinity	Aptevo	Recombinant	OC	OC but APTT reagent variable
n/a ^a	Kcentra ^b	CSL Behring	Plasma-derived prothrombin complex	OC	No recommendation
n/a ^a	Mononine	CSL Behring	Plasma-derived	OC	No recommendation
n/a ^a	Profilnine SD	Grifols	Plasma-derived FIX complex	OC	No recommendation
Nonacog beta pegol	Rebinyln	Novo Nordisk	Recombinant PEGylated	OC	CS or "selected" OC
Nonacog gamma	Rixubis	Shire	Recombinant	OC	OC

Note: For product dosing recommendations, please refer to the package inserts.

CS, chromogenic substrate; FDA, Food and Drug Administration; INN, ; n/a, not available; OC, one-stage clotting.

^a INN (International Nonproprietary Names) is not available for a plasma-derived product.

^b Not indicated for hemophilia B.

TABLE 3 Investigational recombinant FVIII and FIX products

Product	Company	Source/Modification	Reference
DalcA, dalcinonacog alfa	Catalyst	Super-active FIX	25, 63
rFVIII-Fc-VWF-XTEN (BIVV001)	Biogen/Sanofi	VWF-D'D3 domain fused to FVIII via immunoglobulin-G1 Fc domains and 2 XTEN polypeptides	64
SubQ-8	Octapharma	Simoctocog alpha combined with OCTA12, a fragment of the von Willebrand factor	65, 66

FVIII, factor VIII; FIX, factor IX; vWF, von Willebrand factor.

a PEGylated FIX product, which gives an artificially low activity readout in one type of OC assay because of abnormal interactions between the PEGylated product and that particular APTT contact activator.^{23,53}

5.2 | Clinical relevance of the CS and OC assays

It may be tempting to choose a potency assay based on its ability to reflect the true measure of hemostasis potential (i.e., to detect factor deficiency as verified by actual clinical manifestations of the deficiency). For example, it is reported that the CS assay is able to detect some types of mild hemophilia A, whereas the OC assay would report normal values in these samples.^{23,55–59} This is most likely due to the presence of structural and/or functional variants of coagulation factors found in individuals with moderate hemophilia symptoms. Similar to modified recombinant FVIII concentrates, these factor variants are functionally different from wild-type FVIII in the WHO IS, thus rendering different results in the CS and OC assays. However, the observation that the CS assay is more sensitive to some FVIII variants should not be extrapolated to all FVIII variants

because the OC assay was reported to give lower FVIII values than the CS assay in other rare instances.^{23,55–59} Thus, the sensitivity of an assay to a coagulation factor variant is specific to the individual structural and functional characteristics of the modified molecule, whether these modifications are introduced congenitally or through chemical or recombinant means.

The practice of establishing evidence for the effective replacement therapy is centered around demonstrating that increasing factor levels is enough to reduce bleeding episodes, and not an assessment of factor activity by an arbitrary assay. The activity and measurement of factor products, which are structurally modified, are not expected to correlate with the factor variants found in hemophilic patients. Thus, the assays used in the clinical diagnosis of factor deficiency and the behavior of factor products in these assays appear to be unrelated scenarios.

Similarly, the relative physiological relevance of either the OC- and CS-based *in vitro* assays is rather questionable because of how they are designed and performed. Although the OC assay measures the salient hemostatic endpoint of clot formation in plasma, the reaction is initiated by the nonphysiological contact pathway of blood coagulation using artificial reagents (e.g., synthetic lipids).

Conversely, the CS assay is based on a clearly defined mechanism of action using purified components of the coagulation pathway, but it lacks the physiological matrix of patient plasma proteins, which are substantially diluted in this assay.

Another possible criterion for selection is assay robustness. Indeed, the CS assay was originally developed to address the relatively high variability of the OC assay in clinical laboratories in the early 1980s. At the time of its introduction, the CS assay was considered more precise and robust prompting the European Pharmacopeia to include it as a requirement for the potency assignment of FVIII products in Europe. However, advances in clot detection techniques, automation, and reagent stability have resulted in the OC assay performance being comparable to, or showing even better precision and robustness, than the CS assay in routine clinical laboratory practice. Importantly, the OC assay is still more commonly used in clinical laboratories around the world, even though the CS assay has become more readily available in specialized laboratories in hemophilia treatment centers in some countries.¹³

As has been suggested previously,^{51,54,55} both the OC and CS assays should be used throughout product development to investigate potential differences. Both methods are automated, robust, and accurate, and, under the right circumstances, may be suitable for the potency assignment of factor products.

5.3 | Assay harmonization

Harmonization of assays requires a consensus from all stakeholders on the assay methodology to be used, including the manufacturer's methods and those used in clinical laboratories, which has yet to be achieved in the hemophilia field. The World Federation of Hemophilia and the National Hemophilia Foundation recommend the use of both the OC and CS assays.^{1,54} The European Pharmacopeia stipulates the potency of FVIII products be assigned by the CS assay, whereas that of FIX be assigned by the OC assay. The United States Food and Drug Administration (FDA) uses an evidence-based, product-specific approach for new factor products. As a result, some FVIII products approved by the FDA have potency assignments derived from OC assays, but others from CS assays. These differences in regional regulatory requirements for factor products have complicated the effort on assay harmonization.

Harmonization of assay type (i.e., the CS or OC assay) will not in itself assure consistency of potency assignments because of other variables within each methodology (e.g., there are more than 60 different APTT reagents that can yield different test results in the OC assay). In the prescribing information for one recombinant FIX product, differences in assay results up to 40%, caused by the APTT reagent or reference standards used in the assay, were described.⁵⁵ A study comparing extended half-life FIX products to the WHO IS demonstrated differences in potency values, up to a 5-fold overestimation, and a 5-fold underestimation for the same FIX molecule.^{23,53,56}

5.4 | Traceability to the international standard

From a purely biochemical perspective, the structural differences between the bioengineered factor product and endogenous human proteins suggest that the existing factor concentrate IS, which are derived from normal human plasma, may not be the ideal reference for recombinant factor products.⁵⁷ To address this, a panel of product-specific IS, made from their respective recombinant proteins (to ensure a "like versus like" comparison), assigned in IU has been proposed for the development of each factor product.¹¹ The downside of this approach is that an internationally recognized unit of factor activity would have to be assigned to each of these numerous reference materials, which would need to be independently maintained and distributed by the WHO. Alternatively, a panel of product-specific secondary reference materials (e.g., an international reference reagent panel) could be developed such that each international reference reagent is assigned a potency value in the way that is traceable to an existing plasma-derived IS. Either approach would mitigate the issues of assay discrepancy, but would require clinical laboratories to maintain multiple product-specific standards, or to limit testing to patients who receive products that are commutable to the reference used by the laboratory. Patient reporting of specific factor product use would be important to ensure that the clinical laboratories use the specific factor product reference material. The burden of maintaining a "product panel" notwithstanding, this scenario has the theoretical potential to assure accurate dosing for a wide range of hemophilic patients treated with different factor products.

The need for, and feasibility of using, product-specific standards for modified coagulation factor products at clinical laboratories has been discussed at the WHO hearing on the development of product-specific reference materials for coagulation FVIII and FIX products in November 2019 (reviewed in Ovanesov et al¹¹). On the one hand, clinical laboratories are concerned with reporting potentially inaccurate measurements, which can result in incorrect dosing. On the other hand, many participants felt that the use of multiple product-specific factor activity assays is not feasible for a typical clinical laboratory, because of the relatively low volume of testing done for hemophilia indication. A more practical approach was discussed in which a laboratory could use product-specific reference materials to assess the magnitude of bias of the in-house assay, but not for the generation of a calibration curve in each assay.¹¹ In cases of substantial bias, the laboratory should inform the clinicians about this bias and the products that are not compatible with the reagents and assays used by this particular laboratory.

5.5 | Meaning of the international unit

The lack of commutability between factor products and the current WHO IS, as manifested in assay discrepancies, may prompt us to re-examine the definition of the IU as the amount of FVIII or FIX in 1 ml

of pooled normal plasma, and its applicability to recombinant factors that are different in structure and function to the IS.⁵⁷ For example, there are no PEGylated molecules or Fc-fusion molecules in normal plasma, which previously have led to a question as to why products like PEGylated FVIII and Fc-fused FVIII should be calibrated in FVIII IU at all.⁵⁷ As it stands, all currently licensed factor products remain labeled in IU traceable to its respective WHO IS. Thus, the IU still finds its utility in the hemophilia community.

One exception is a recombinant porcine FVIII product labeled with its own product-specific unit (a non-IU), but even this activity unit is traceable to an IS, as explained by the manufacturer.⁵⁸ The use of a product-specific unit has not resulted in reports of serious safety or efficacy concerns likely because the porcine FVIII product is indicated for the treatment of acquired FVIII deficiency, and the way it is dosed does not follow commonly accepted, FVIII level-dependent dosing regimens typical of congenital hemophilia using recombinant human coagulation factor concentrates labeled in IU.

5.6 | Standards of treatment

Thus far, the practice of labeling potencies for factor products in IU is understood by both patients and clinicians as an integral part of hemophilia management in establishing a standard dosing recommendation. Indeed, treatment of minor bleeding episodes by most licensed FVIII products is dosed at approximately 10 to 20 IU/kg, which is validated by their respective single-arm (i.e., the efficacy of a single product) clinical trials. However, because these were not comparative studies, the similarity in dosing recommendations for these products should not be construed as equivalency or interchangeability of products. That is to say, clinical outcomes are to be closely monitored should the individual decide to switch to a new product. Likewise, the suitability of one assay type for one product may not be extrapolated to similar products. Nonetheless, the clinician's overall knowledge of both the products that are administered to patients and the assays used to assess a patient's response to said products is pivotal in providing the best care for the patient. Whether clinicians should limit the use of products to those that can be measured accurately by local laboratories falls under the purview of the practice of medicine, which is dependent on the clinicians' best judgment based on their experience and knowledge of the patient, and the product and its various characteristics, including its behavior in different activity assays.

The consensus dosing range can be wide. As outlined in the third edition of the World Federation of Hemophilia Guidelines for the Management of Hemophilia, conventional prophylaxis with standard half-life FVIII products is divided, according to its intensity, into a high-dose regimen at 25 to 40 IU FVIII/kg every 2 days, an intermediate-dose regimen at 15 to 25 IU FVIII/kg 3 days per week, and a low-dose regimen at 10 to 15 IU FVIII/kg 2 to 3 days per week (i.e., the range within each dosing regimen is broad enough to accommodate most assay discrepancies).¹

5.7 | Considerations for patient safety

The lack of assay agreement and harmonization can have undesirable consequences on patient health. Differences in potency assignments by manufacturers and factor recovery values in clinical laboratories can lead to wrong decisions and under- or overdosing of the patient, resulting in bleeding or potential thrombotic complications, respectively. Thus, during product development and in clinical field tests, it is desirable to use both assays to demonstrate how the potency results may influence the assessment of factor activity in blood.^{51,59,66}

5.8 | Clinical validation

The safe and effective dosing of any therapeutic agent is validated in clinical studies. In factor products, dosing and postadministration monitoring are especially important because of how these products are traditionally used. During the development of new factor concentrates, much effort has been directed to determining the appropriate assay(s) for factor potency assignment and product monitoring. Product labeling for licensed products relays evidence-based recommendations on the appropriate methods that should be used by clinical laboratories (see Tables 1 and 2). Thus, the need to present information in product labeling underscores the importance of testing new factor products with both CS and OC assays using different reagents during product development (e.g., in clinical laboratory studies), and validating the assay readouts with clinical outcomes (in efficacy studies) to prevent dosing errors.

5.9 | Clinical laboratory field studies

To aid in the postadministration monitoring of activity levels in patients, a comparative field study with factor-deficient plasma samples spiked with the specific factor product can also provide valuable information on the scope of real-world assay differences facing clinical laboratories. In these multicenter studies, each laboratory uses its local assays to test the spiked plasma samples and compares the results with those from other participating laboratories to evaluate the magnitude of the discrepancies.⁴⁴⁻⁵⁰ The information, which can be included in the product package insert, will be especially useful for clinical laboratories in monitoring product levels in patient plasma postadministration since the methodology, reagents, and reference standards used in clinical laboratories are most likely different from those used by the manufacturer for potency assignment of the product.

In one example, when discrepancies in results between the recommended and alternative assays are consistent and reproducible within the expected range of the method variability, product labeling recommends a mathematical conversion step to determine the "actual" effective level of factor activity in the patient (see Table 1, lonoctocog alfa). Of note, the potency-assigned samples provided

by the manufacturer of licensed products can be useful in maintaining the external quality assurance by clinical laboratories involved in routine testing of plasma samples from individuals with hemophilia dosed with the product.

The clinical laboratory studies are critical in the understanding of issues related to postinfusion monitoring, and they may also inform product manufacturers on issues related to the potency assignment. However, data from clinical laboratories would not resolve the discrepancy in potency assays of the factor concentrate. Indeed, one should be cognizant of the differences between the clinical laboratory and the manufacturer's potency assignment exercises, such as differences in the reference standards used (pooled plasma for the clinics and purified concentrate for manufacturers) and sample dilution (low factor activity in hemophilia plasma and high concentration in product vials).

6 | ASSAY DISCREPANCIES IN GENE THERAPY

Another area where an awareness of factor activity assay discrepancies may be important is gene therapy for hemophilia, which is being developed as a single-dose treatment that may provide long-term expression of the deficient coagulation factor in patients at a level that can reduce or eliminate the need for frequent administrations of factor products. Indeed, similar to recombinant factor concentrates, the coagulation factors expressed after gene therapy (e.g., BDD-FVIII or the hyperactive Padua variant of FIX) may be functionally and/or structurally different from their endogenous counterparts, which would also lead to discrepancies in results when assayed using different factor activity methods.^{25,59,60} Consequently, some of the considerations in this review may also be applicable to gene therapy studies. For example, the recommendation on performing field studies with clinical laboratories has been described in a recently published FDA Guidance for Industry on Human Gene Therapy for Hemophilia.⁶¹

Interestingly, recent evidence indicates that the factor proteins expressed in gene therapy patients and in mammalian cell substrates may show substantially discrepant activities, even when both modalities use the same transgene, such as an SQ-variant of BDD-FVIII.⁶⁰ In recombinant BDD-FVIII products, the CS assay usually reports a higher value than the OC assay.^{32,34} In contrast, higher OC values than CS values were recently reported in FVIII activity in hemophilia A individuals who have undergone gene therapy.⁶⁰ Thus, CS and OC assay discrepancies have again complicated the interpretation of factor activity measurements and the assessment of the effectiveness of gene therapy in these individuals.

7 | CONCLUSIONS

The development of modified recombinant FVIII and FIX products has yielded promising improvements in the treatment of individuals with hemophilia. Further, assignment of potency in IU, referencing

the relevant WHO IS, would maintain the traditional practice of how hemophilia is managed safely and effectively by regimens familiar to clinicians and patients. However, discrepancies in potency values derived from OC and CS assays, which stem from the distinctive structural and functional properties of these modified products, may lead to bleeding or a potential risk of thrombosis if these products are not dosed properly. These assay disagreements are also observed in other treatment modalities for hemophilia, such as coagulation factor mimetics and gene therapy.

With a diversity of preferences among stakeholders (i.e., factor product manufacturers, clinical laboratories, and regulatory agencies), the rationale behind the selection of an assay for potency labeling and product monitoring is not always well explained. Furthermore, some of the common arguments (e.g., analytical or physiological advantage of one assay system over another) do not withstand a critical inspection and may carry little weight in the regulatory decision-making process. Thus, we have presented here a rational scientific review of approaches that have been previously suggested for choosing a suitable assay for assigning potency and factor level monitoring.

In summary, both OC and CS assays are usually used to assess product activity and tested for commutability (parallelism) with the current WHO IS during all stages of product development.⁵¹ Furthermore, clinical laboratory field studies are performed to ensure proper evaluation of the magnitude of potential discrepancies.⁵¹ Providing as much information as possible in the product package insert and relevant educational materials is also helpful to ensure the safe and effective use of the product.¹¹ For a new, modified product, readily available technical support and product-specific reference materials aid in the accurate assessment of its recovery levels in patients.¹¹ Ultimately, the assays are evaluated based on method validity, patient safety, as well as clinical relevance. Under these conditions, there appears to be a consensus within the hemophilia community to support the labeling of new factor products in IU to maintain dosing tradition and understanding among clinicians, patients, and other stakeholders.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest and state that they had no interests that might be perceived as posing a conflict or bias.

AUTHOR CONTRIBUTIONS

Mikhail V. Ovanesov and Joseph W. Jackson wrote the paper with help and input from Basil Golding and Timothy K. Lee.

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